PLAIN-LANGUAGE TECHNICAL REPORT

# Four AI agents and the complete model inventory

How CataTriage-AI processes information, what each saved model can do, and what every reported score really means.

**IMPORTANT** This is an educational research system. It does not diagnose, treat, assign clinical urgency, or replace a doctor.

## The four-agent workflow

| STEP | AGENT | WHAT IT DOES | OUTPUT | SAFETY |
|---|---|---|---|---|
| **01** | **Data Preprocessing** Patient fields and an image | Checks missing fields, image format, size and basic technical quality. | Clean input plus quality notes | Stops corrupt or incomplete input. |
| **02** | **Model Inference** Validated input from Agent 1 | Runs only the model allowed for that data type and records probabilities. | Model output or not_available | Never turns a score into urgency. |
| **03** | **Safety and Rule Validation** Model output and scope information | Checks modality, model availability, uncertainty and fixed safety rules. | Flags and clinician-review requirement | Blocks unsupported clinical claims. |
| **04** | **Explanation and Reporting** Validated evidence from Agents 1-3 | Writes an easy summary, limitations and an auditable report. | Doctor-facing explanation | No diagnosis or treatment advice. |

**FLOW** Preprocessing > inference > safety validation > explanation and reporting. Every stored trace ends with clinical_priority = null and requires_clinician_review = true.

## What happens for a real slit-lamp upload?

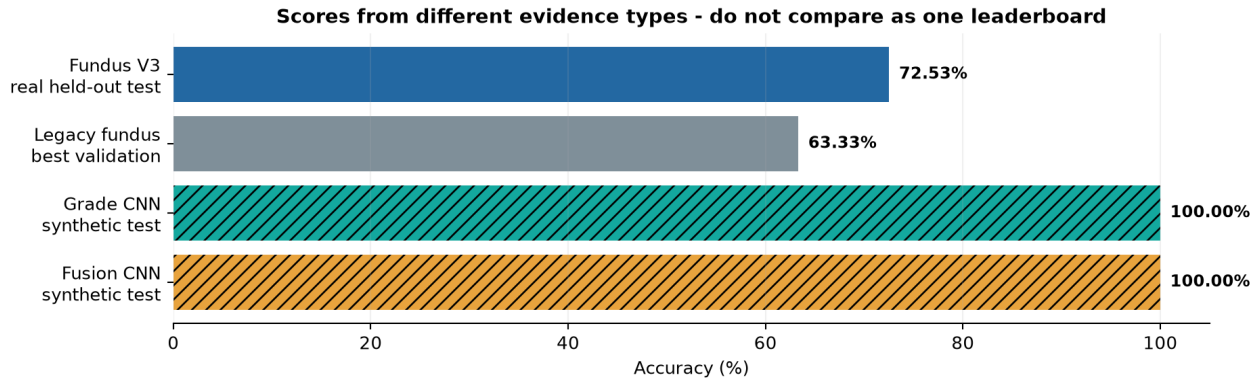
Agent 1 checks the file. Agent 2 returns not_available because no validated real slit-lamp grade model exists. Agent 3 explains that this is not a negative result and requires a doctor. Agent 4 creates the report. The administrator sees only a tier that a doctor entered manually.

MODEL INVENTORY

# Every model and its evidence

Scores below come from different datasets and answer different questions. Synthetic scores must never be compared with real-image scores as if they were clinical evidence.

| MODEL | STATUS | DATA | ACCURACY | MACRO F1 | EVALUATION | PLAIN MEANING |
|---|---|---|---|---|---|---|
| Fundus EfficientNet-B0 V3 | Trained / evaluated | Real dataset folder labels | 72.53% test | 66.03% test | 91 test images | Four-class fundus baseline; not cataract grading. |
| Legacy fundus EfficientNet-B0 | Preserved checkpoint | Real dataset folder labels | 63.33% validation | 60.15% validation | 90 validation images | Earlier model; no separate final-test metric recorded here. |
| Synthetic cataract grade CNN | Trained / education | Generated lens patterns | 100.00% test | 100.00% test | 800 synthetic tests | Learns four procedural grades; not human results. |
| Synthetic clinical fusion CNN | Trained / education | Generated image + 6 fields | 100.00% test | 100.00% test | 800 synthetic pairs | Shows CNN and structured-data fusion. |
| Real multi-data cataract grade | Not trained | Real slit-lamp + clinical | Not available | Not available | 0 linked labels | Blocked: labels and image-record crosswalk missing. |
| Real clinical fusion | Not trained | Real paired multimodal data | Not available | Not available | 0 verified pairs | Blocked: paired cohort, target and external validation missing. |

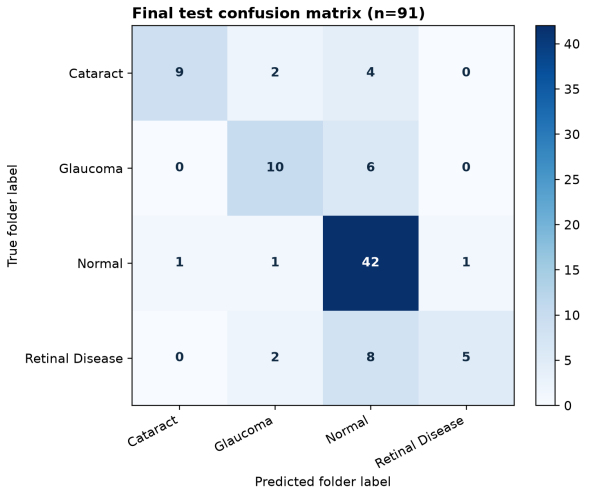


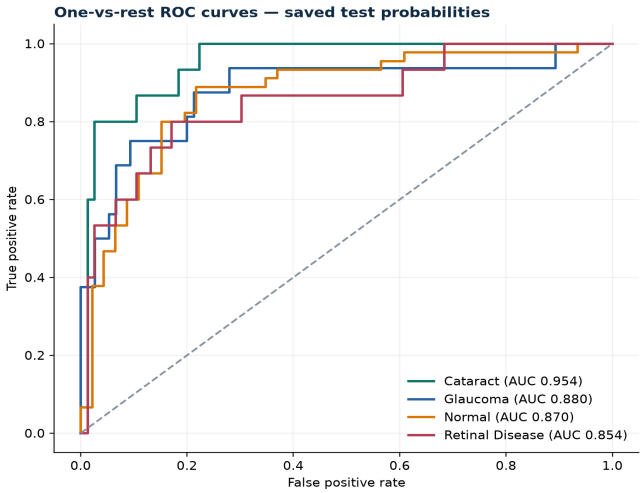

Striped bars are synthetic-only results. The legacy fundus value is validation accuracy, while V3 uses a held-out test. These bars are included for inventory, not direct ranking.

**BEST REAL RESULT** Fundus V3: 72.5275% accuracy, 66.0338% Macro-F1 and 88.9266% Macro one-vs-rest ROC-AUC on 91 held-out images.

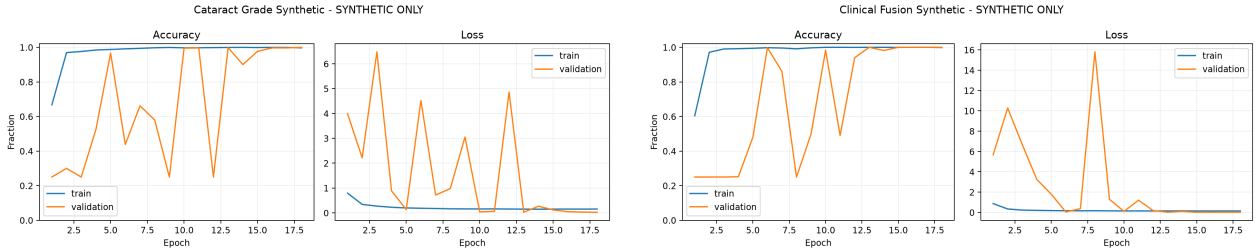
GRAPHS AND EASY INTERPRETATION

# What the results show




Real fundus V3: the confusion matrix shows 66 correct images out of 91. Normal images were easiest (42 of 45 correct). Retinal disease was hardest (5 of 15 correct). The ROC curves show separation across thresholds, but the test is small and image-level.




Synthetic training: both models eventually reached perfect validation and test scores, although early validation moved up and down. The saved best checkpoints are epoch 18 for the image CNN and epoch 13 for the fusion CNN. Training used the RTX 2050 and took about 103 seconds and 83 seconds respectively.

## Simple definitions

| | |
|---|---|
| **Accuracy** | The percentage of all examples classified correctly. |
| **Balanced accuracy** | Accuracy averaged across classes, so large classes do not dominate. |
| **Macro F1** | A combined precision-and-recall score calculated equally for every class. |
| **ROC-AUC** | How well a model separates a class across many probability thresholds. |

| | |
|---|---|
| **BOTTOM LINE** | The application, GPU training, audit trail and four-agent orchestration work. The only real held-out model result is the fundus V3 baseline. The two perfect scores are useful demonstrations, but real cataract grading still requires verified image-linked clinician labels and external validation. |